2.1 Consider the following five observations. You are to do all the parts of this exercise using only a calculator.

x	y	$x-\bar{x}$	$(x-\bar{x})^2$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
–1	1	–2	4	–1	2
0	0	–1	1	–2	2
$\sum x_i = 5$	$\sum y_i = 10$	$\sum(x_i-\bar{x}) = 0$	$\sum(x_i-\bar{x})^2 = 10$	$\sum(y_i-\bar{y}) = 0$	$\sum(x_i-\bar{x})(y_i-\bar{y}) = 8$

(a) $\bar{x} = 1,\ \bar{y} = 2$

(b) $b_2 = \dfrac{\sum(x-\bar{x})(y-\bar{y})}{\sum(x-\bar{x})^2} = \dfrac{8}{10} = 0.8$

$b_1 = \bar{y} - b_2\bar{x} = 2 - 0.8\times 1 = 1.2$

(c) $\sum_{i=1}^{5} x_i^2 = 3^2+2^2+1^2+(-1)^2+0^2 = 15$

$\sum_{i=1}^{5} x_i y_i = 3\times4+2\times2+1\times3+(-1)\times1+0\times0 = 18$

$\sum_{i=1}^{5} x_i^2 - N\bar{x}^2 = 15 - 5\times1^2 = 10 = \sum_{i=1}^{5}(x_i-\bar{x})^2$

$\sum_{i=1}^{5} x_i y_i - N\bar{x}\bar{y} = 18 - 5\times1\times2 = 8 = \sum_{i=1}^{5}(x_i-\bar{x})(y_i-\bar{y})$

(d) $s_y^2 = \dfrac{10}{4} = 2.5$ $\quad s_{xy} = \dfrac{8}{4} = 2$

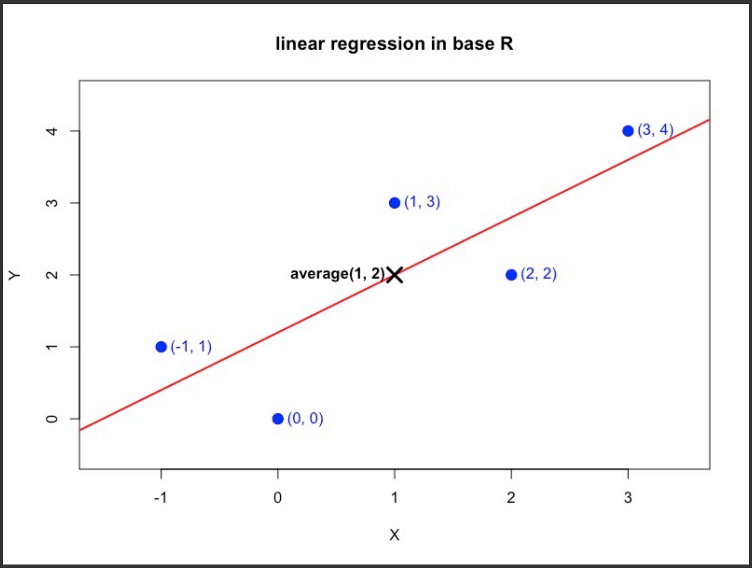
$s_x^2 = \dfrac{10}{4} = 2.5$ $\quad r_{xy} = \dfrac{s_{xy}}{s_x s_y} = \dfrac{2}{\sqrt{2.5}\sqrt{2.5}} = 0.8$

$CV_x = 100\left(\dfrac{s_x}{\bar{x}}\right) = \dfrac{100\sqrt{2.5}}{1} = 50\sqrt{10}$

median $(x) = 1$

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i\hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
–1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i = 5$	$\sum y_i = 10$	$\sum \hat{y}_i = 10$	$\sum \hat{e}_i = 0$	$\sum \hat{e}_i^2 = 3.6$	$\sum x_i\hat{e}_i = 0$

(e) (f)



(g) $b_1 = 1.2$

$b_2 = 0.8$

$\bar{x} = 1$

$\bar{y} = 2$

$\bar{y} = b_1 + b_2\bar{x} \Rightarrow \bar{y} = 2 = b_1 + b_2\bar{x} = 1.2 + 0.8(1) = 2$

(h) $\bar{\hat{y}} = \sum_{i=1}^{5} \frac{\hat{y}_i}{N} = \frac{3.6+2.8+2+0.4+1.2}{5} = 2 = \bar{y}$

(i) $\hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} = \frac{3.6}{3} = 1.2$

(j) $var(b_2) = \frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2} = \frac{1.2}{10} = 0.12$

$se(b_2) = \sqrt{var(b_2)} = \sqrt{0.12} = \frac{\sqrt{3}}{5}$